Transfer: momentum and impulse

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Apply momentum reasoning to new collisions and deflection cases.

SUCCESS CRITERIA - I can...

1. Define the system and positive direction.
2. Use conservation or impulse only when assumptions apply.
3. Explain energy and model limits clearly.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “momentum” and “impulse” to explain your model or result. Sentence starters are optional.

momentum: mass times velocity, including direction

impulse: net force acting over time; change in momentum

conserved: unchanged in total under the stated conditions

01 Warm-up: bring back an earlier idea

In the same straight-line deflection model, keep sideways Δv fixed but make lead time 3 times as long. **Compare** shift and name a limitation.

Word help: deflection = a change in the direction or path of motion

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

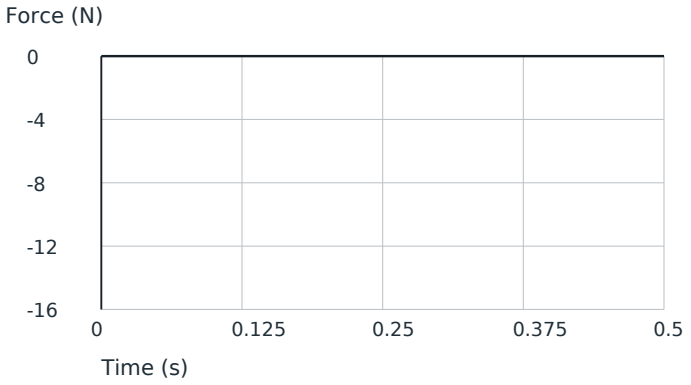
02 Worked example: explain the model

Reference: $p = mv$; $J = \Delta p = F_{avg}\,\Delta t$; $p_{total,\,i} = p_{total,\,f}$.
Use momentum conservation only when external impulse is negligible. Label before/after diagrams and directions.

Worked reasoning: A 2 kg cart moving at 3 m/s stops in 0.5 s. Momentum change is -6 kg·m/s; average force is -12 N. Force direction opposes the initial motion.

Your turn: Represent the same stop with momentum arrows and a force-time rectangle.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

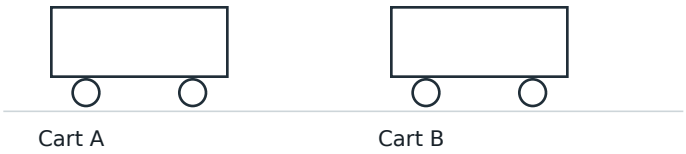
03 Read and interpret the model

A 1 kg cart at 4 m/s right sticks to a 1 kg cart at rest.
Draw before/after and explain which quantity is conserved.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: My claim is … . The diagram/data show …, so … .

Before contact: add velocity arrows.



Marzano check - circle: ① ② ③

04 Apply the physics

A 2 kg cart moving right at 6 m/s stops in 5 s.
Calculate its momentum-change magnitude and average stopping-force magnitude.
Use the before-and-after model to show the direction of the change.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.

Before	**After**

Label the initial and final momentum.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A report uses equal total momentum before/after to claim a collision is elastic. What additional calculation is required?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A 4 kg cart travels left at 6 m/s.
Find momentum and draw its direction.
Use before and after snapshots to organize the momentum account.

Sentence starter: I found … [unit]. This value means … in this situation.

Before	**After**

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

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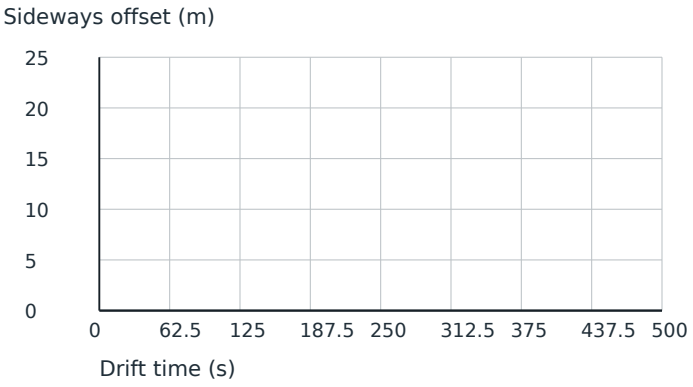
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A fixed impulse gives a target 0.04 m/s sideways velocity change.
Find shift after 500 s and state a model limitation.

Sentence starter: My claim is … . The diagram/data show …, so … .

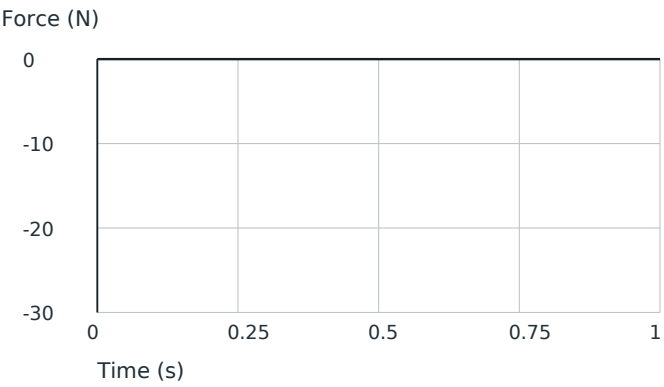


Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2.
Design two stopping processes for this cart that give the same impulse but different average forces.
Explain the time-force tradeoff.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3